University of Hull

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GeoMentor: A Hybrid IRT/BKT Intelligent Tutoring System for UK Geography Education

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Abstract

This project designs, implements, and evaluates GeoMentor: a production-deployed adaptive intelligent tutoring system (ITS) for UK KS3/GCSE Geography. The system implements a hybrid Item Response Theory 3-Parameter Logistic (IRT 3PL) and Bayesian Knowledge Tracing (BKT) algorithm that continuously estimates learner ability via Expected A Posteriori (EAP) updating, tracks per-Knowledge Component mastery state, and selects items targeting the learner's Zone of Proximal Development using Fisher information maximisation. A novel credit-factor modification to the standard BKT learning transition equation is introduced, weighting posterior updates according to the maximum hint level accessed per question. Bloom's taxonomy is implemented as a computational selection constraint, enforcing cognitive prerequisite ordering across a 13-KC curriculum graph aligned to the AQA GCSE Geography specification.

The project follows a Design Science Research methodology across ten two-week development sprints. Evaluation adopts a five-track multi-method validity framework of algorithm validation, engagement data, System Usability Scale (SUS) usability, misconception analysis, and system robustness monitoring, adopted in response to the constraint of $n = 4$ external participants, for which inferential statistics are not viable.

All four participants exhibited positive within-session $\Delta\hat{\theta}$ (range +0.379 to +0.571 logits; mean +0.479) at 80–87% response accuracy across 15-question adaptive sessions, consistent with Zone of Proximal Development targeting. Two participants who completed the in-app System Usability Scale scored 92.5 and 87.5 (mean 90.0), exceeding the ≥ 85 "excellent" threshold. The system achieved 99.9% uptime with zero session state corruption events over the evaluation period.

Five concrete contributions are made: (1) the first publicly accessible adaptive ITS for UK Geography; (2) a credit-factor BKT modification formalising scaffolding usage as graded mastery evidence; (3) Bloom-gated adaptive progression as a live selection constraint; (4) a five-track multi-method evaluation framework reusable for small-sample ITS validation; and (5) a domain-agnostic open-source ITS architecture parameterised by KC identifiers

and IRT parameter files, requiring no algorithmic changes to support additional subject domains.

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Chapter 1

Introduction

1.1 Background and Motivation

The dominant paradigm in digital learning platforms remains one-size-fits-all content delivery. A learner who already understands tectonic plate mechanics receives the same sequence of questions as one encountering the concept for the first time. This homogeneity is not a technical constraint but a design choice that the empirical literature consistently identifies as suboptimal.

Bloom (1984) demonstrated that one-to-one human tutoring, which adapts to the learner's current ability in real time, produces learning outcomes approximately two standard deviations above conventional classroom instruction. VanLehn (2011) subsequently showed that step-based Intelligent Tutoring Systems (ITS) can approach this performance at scale, achieving effect sizes statistically indistinguishable from human tutoring. The adaptive feedback loop is the critical variable, not the delivery medium.

Despite this evidence, adaptive ITS remain largely confined to mathematics and programming domains, where knowledge structures are well-defined and assessment is unambiguous. Geography, integrating declarative factual knowledge, procedural reasoning, and conceptual understanding, has received almost no attention in the adaptive learning literature (Bednarz et al. 2013). The UK KS3/GCSE Geography curriculum represents both an underserved educational context and a meaningful test case for ITS methods applied to non-STEM knowledge domains.

1.2 Research Question

This project addresses the following research question:

> *Can a hybrid IRT 3PL + BKT adaptive engine, implemented in a production web application, deliver personalised question sequencing that produces measurable within-session learning gains for learners studying UK KS3/GCSE Geography content?*

The question has three components: an *algorithmic claim* (the hybrid IRT+BKT engine can perform valid adaptive selection), a *technical claim* (implementable in a production-deployed web application), and an *empirical claim* (the system produces measurable learning gains). The evaluation in Chapter 5 addresses all three, with the acknowledgement that the empirical claim is evaluated at proof-of-concept scale rather than at statistical significance.

1.3 Project Overview

GeoMentor is an adaptive intelligent tutoring system for UK KS3/GCSE Geography, accessible at `https://e-learning-final-project.vercel.app`. The system implements a hybrid IRT 3PL + BKT algorithm that continuously estimates learner ability and Knowledge Component mastery, selects questions targeting the learner's Zone of Proximal Development, and enforces Bloom's taxonomy cognitive prerequisites across a 13-KC curriculum graph.

The system was developed over ten two-week sprints from October 2025 to May 2026, following a Design Science Research methodology (Hevner et al. 2004). It is production-deployed, publicly accessible, and open-source, providing a proof-of-concept implementation that can be independently verified and extended.

1.4 Contributions

This project makes the following contributions:

1. **A production-deployed hybrid IRT 3PL + BKT adaptive engine for UK Geography.** The first publicly accessible ITS targeting the UK KS3/GCSE Geography curriculum, extending the empirical evidence base for adaptive learning beyond traditional STEM domains.

2. **A credit-factor modification to the standard BKT update.** A novel modification of the BKT learning transition equation that weights correct responses by a graduated credit factor reflecting the maximum hint level accessed, formalising the evidential relationship between scaffolding usage and mastery evidence.

3. **Bloom-gated adaptive progression.** An implementation of Bloom's taxonomy as a computational selection constraint, enforcing that learners consolidate recall-level knowledge before encountering application-level items.

4. **A five-track multi-method evaluation framework for small-sample ITS validation.** A structured evaluation approach combining algorithmic validation, engagement data, usability metrics, misconception analysis, and robustness monitoring, designed for contexts where inferential statistics are not viable due to small sample sizes.

5. **An open-source domain-agnostic ITS architecture.** The codebase provides a reusable template for researchers and developers building adaptive systems for non-STEM declarative-knowledge domains.

1.5 Scope and Delimitations

- **Domain.** UK KS3/GCSE Geography only (AQA specification).
- **Assessment type.** Multiple-choice questions with four options.
- **Participant population.** University of Hull students (convenience sample); the system targets KS3/GCSE students but was evaluated with adults due to ethical and logistical constraints.
- **Algorithm.** IRT 3PL + BKT hybrid. DKT, Elo, and 2PL were evaluated and rejected (Section 2.2).
- **Evaluation scale.** Proof-of-concept at $n = 4$. No inferential statistical claims are made.

1.6 Ethical and Professional Considerations

All participant data was collected under informed written consent. No personally identifiable data beyond email address and name (required for authentication) was collected; session performance data is stored under a pseudonymous user ID, accessible only to the authenticated user and to the researcher under the study protocol.

AI use declaration. This project was developed with AI assistance (Claude, Anthropic), consistent with the University of Hull's Student AI Use Policy (2025). AI assistance was employed for infrastructure scaffolding (Supabase RLS boilerplate, Resend email integration, bcrypt helpers), frontend component implementation from candidate-authored specifications, Mapbox integration, Vercel deployment configuration, and Prisma schema iteration. The following constitute the candidate's independent intellectual work and were not produced by AI tools: the hybrid IRT 3PL + BKT algorithm architecture; the credit-factor BKT modification and its full mathematical derivation; the Bloom escalation gating design and its implementation as a computational constraint; the 13-KC curriculum ontology and prerequisite DAG; the research design including the within-subjects quasi-experimental structure, VanLehn benchmark selection, and Sprint 5 scope reduction justification; the five-track multi-method evaluation framework; all Architecture Decision Records; and the IRT calibration strategy. AI assistance was employed exclusively as an implementation accelerator for repetitive or infrastructural tasks; all design decisions and algorithmic contributions are the candidate's own work and are grounded in the peer-reviewed literature cited throughout this report.

Data collection complied with UK GDPR and the Data Protection Act 2018. The lawful basis for processing is *consent*: participants provided explicit informed written consent prior to any data collection and were informed of their unconditional right to withdraw without consequence. No special category data were collected; all data are stored on EU-resident servers (Supabase, Frankfurt region). This project was conducted in accordance with the British Computer Society Code of Conduct's four obligations (public interest, professional competence, integrity, and duty to relevant authority); ethical considerations were assessed under the University of Hull Research Ethics Policy prior to participant recruitment.

1.7 Report Structure

Chapter 2 critically reviews adaptive ITS theory, algorithms, CAT principles, empirical effectiveness, and ethical dimensions, identifying five research gaps. Chapter 3 details the KC ontology, adaptive algorithm design with full mathematical derivation, evaluation protocol, and agile development methodology. Chapter 4 describes the technical implementation including database schema, adaptive engine, API routes, and security. Chapter 5 presents the five-track multi-method evaluation. Chapter 6 interprets results, examines limitations, and identifies future work. Chapter 7 summarises contributions and offers a final statement.

Chapter 2

Literature Review

2.1 Theoretical Foundations of Adaptive Learning

2.1.1 The 2-Sigma Problem and ITS

Bloom (1984) demonstrated that one-to-one human tutoring produces learning outcomes two standard deviations above conventional instruction (the "2-sigma problem"). VanLehn (2011) partially resolves this at scale: step-based ITS achieves $d = 0.76$, statistically indistinguishable from human tutoring ($d = 0.79$). The critical determinant is interaction granularity, not algorithmic sophistication—a finding that directly informs the scaffolding architecture of this system. Chi et al. (2008) provide a complementary account: the effectiveness of human tutors derives less from domain expertise and more from eliciting and responding to student reasoning, suggesting that ITS design should prioritise responsive feedback over predictive accuracy.

2.1.2 Zone of Proximal Development

Vygotsky's (1978) Zone of Proximal Development (ZPD) defines the cognitive space between independent and scaffolded achievement. Tasks below the ZPD produce boredom; tasks above it produce cognitive overload; only tasks within it produce optimal learning. This system operationalises the ZPD through IRT target probability $P(\theta) \approx 0.70$: the selected item presents difficulty at which the learner has approximately 70% probability of success. The 13-KC prerequisite graph encodes the multi-dimensional knowledge structure required to navigate Geography's integration of declarative, procedural, and conceptual knowledge (Park and Lee 2004).

2.1.3 Mastery Learning and Bloom's Taxonomy

Bloom (1984) proposed mastery learning: progression is gated by demonstrated competence rather than time-on-task. Ritter et al. (2016) identify 80% accuracy as the optimal threshold from large-scale Cognitive Tutor data; this system implements $p(\text{Learned}) \geq 0.80$ per KC as the BKT mastery gate. Bloom's revised taxonomy (Anderson and Krathwohl 2001) provides three cognitive levels (Remembering, Understanding, Applying); the adaptive engine enforces escalation gates at $p(\text{Learned}) \geq 0.40$ (L1→L2) and ≥ 0.60 (L2→L3), ensuring declarative knowledge is consolidated before analytical demands arise.

2.1.4 Spaced Repetition and Cognitive Load

Cepeda et al. (2006) establish the spacing effect: distributed practice consistently outperforms massed practice at delayed retention. This system implements Successive Relearning (Kornell and Bjork 2009) through BKT decay detection: KCs whose $p(\text{Learned})$ has declined receive a 1.5× information boost at next selection, prioritising re-presentation of consolidating material. Sweller (1988)'s cognitive load theory informs the graduated hint system: each hint level reduces intrinsic load while also reducing the evidential weight of a subsequent correct response (Koedinger and Aleven 2007), a relationship formalised by the credit-factor BKT modification.

2.2 Adaptive Algorithms: A Critical Comparative Analysis

Matos et al. (2025) identify three dominant ITS algorithm families: Elo-based rating systems, probabilistic models (IRT and BKT), and deep learning approaches. Each is evaluated below against four project requirements: per-KC mastery tracking, prerequisite enforcement, item discrimination modelling, and interpretability.

2.2.1 Elo Rating System

Originally developed for chess player ranking (Elo 1978), the Elo system updates learner and item ratings after each response:

$$R_i^{\text{new}} = R_i + K \cdot (\text{outcome} - P(\text{correct})) \tag{2.1}$$

Pelánek (2016) reports approximately 92% prediction accuracy after 25–30 interactions.

However, Elo operates as a single global ability estimate: it cannot model per-KC mastery states, cannot enforce prerequisite dependencies, and provides no knowledge decay detection. These three limitations make Elo unsuitable for a 13-KC curriculum with prerequisite structure, regardless of its calibration simplicity.

2.2.2 Item Response Theory (3PL)

IRT (Embretson and Reise 2000) models response probability as a function of latent ability. The three-parameter logistic (3PL) model (Birnbaum 1968) is:

$$P(X = 1 \mid \theta) = c + \frac{1 - c}{1 + e^{-a(\theta - b)}} \tag{2.2}$$

where a is item discrimination, b is difficulty, and c is the pseudo-guessing parameter (fixed at 0.25 for four-option items). The guessing parameter is critical: Geography MCQ items carry a non-trivial guessing probability, and ignoring it systematically overestimates learner ability following correct responses. The principal limitation is calibration data requirements: traditional 3PL estimation requires 500+ observations per item (Lord 1980), infeasible at student-project scale. This project mitigates by fixing $c = 0.25$ as an equality constraint and pre-specifying a Rasch (1PL) fallback if RMSEA > 0.10 (Chalmers 2012). IRT has no native mechanism for per-KC longitudinal tracking, requiring combination with a knowledge state model.

2.2.3 Bayesian Knowledge Tracing

BKT (Corbett and Anderson 1995) conceptualises learning as a hidden Markov process with binary latent states: a learner either knows or does not know a KC. Four parameters govern each KC: $p(L_0)$ (prior knowledge), $p(T)$ (learning transition), $p(S)$ (slip), and $p(G)$ (guess). After each observation the posterior updates via Bayes' theorem:

$$p(L_n \mid \text{obs}) = \frac{P(\text{obs} \mid L_n) \cdot p(L_n)}{P(\text{obs} \mid L_n) \cdot p(L_n) + P(\text{obs} \mid \neg L_n) \cdot (1 - p(L_n))} \tag{2.3}$$

with learning transition:

$$p(L_{n+1}) = p(L_n \mid \text{obs}) + \big(1 - p(L_n \mid \text{obs})\big) \cdot p(T) \tag{2.4}$$

BKT's critical strength for this project is per-KC resolution, enabling mastery-gated progression. Beck and Chang (2007) identify a parameter identifiability problem for sparse data; this project mitigates it with informed priors and domain-standard $p(S)/p(G)$ values following Pardos and Heffernan (2011).

Credit factor modification. Standard BKT treats all correct responses as equally informative. Koedinger and Aleven (2007) demonstrate that correct responses following worked examples are substantially less informative than unaided correct responses. This project modifies the learning transition by credit factor $\mathrm{cf} \in \{1.0, 0.9, 0.7, 0.5\}$ depending on scaffolding level accessed:

$$p(L_{n+1})^* = p(L_n \mid \mathrm{obs}) + \Big(1 - p(L_n \mid \mathrm{obs})\Big) \cdot p(T) \cdot \mathrm{cf} \tag{2.5}$$

2.2.4 Deep Knowledge Tracing

Piech et al. (2015) introduced DKT, replacing BKT's probabilistic model with an LSTM network achieving AUC improvements of approximately 6% over BKT. Khajah et al. (2016) demonstrate that this advantage largely disappears when BKT is extended with temporal features. More critically, DKT sacrifices interpretability entirely: hidden state vectors carry no psychologically meaningful correspondence, and adaptive decisions cannot be traced to specific knowledge estimates. Matos et al. (2025) confirm that in small-scale educational deployments, interpretable probabilistic systems remain more appropriate. DKT is unsuitable for a project requiring full accountability of every adaptive decision.

2.2.5 Hybrid IRT + BKT: Rationale and Prior Work

The hybrid approach integrates IRT's item-level psychometric precision with BKT's per-KC knowledge tracking. Each algorithm addresses the other's primary limitation: BKT provides no item discrimination; IRT provides no per-KC mastery tracking. Käser et al. (2017) confirm across 1.2 million student–problem interactions that the value of hybrid architectures lies not in marginal accuracy improvement but in simultaneous modelling of item properties and temporal knowledge dynamics. Desmarais and Baker (2012) review 40 learner modelling systems and conclude that hybrid probabilistic approaches consistently outperform single-algorithm systems in structured multi-KC curricula. Table 2.1 summarises the algorithm selection rationale.

Table 2.1: Comparative evaluation of adaptive algorithms against project requirements.

Criterion	**Elo**	**IRT 3PL**	**BKT**	**IRT+BKT Hybrid**
Data requirements	Low	High	Medium	Medium (with priors)
Per-KC mastery tracking	No	No	Yes	Yes
Prerequisite enforcement	No	No	Yes	Yes
Item discrimination modelling	No	Yes	No	Yes
Knowledge decay detection	No	No	Partial	Yes (via decay flag)
Interpretability	High	High	High	High
Calibration stability (small n)	High	Low	Medium	Medium

2.3 Computerised Adaptive Testing

CAT applies IRT to sequential item selection: each successive item is chosen based on all previous responses rather than a fixed predetermined sequence (Wainer et al. 2000). The foundational principle is maximum Fisher information selection (Weiss 1982):

$$I(\theta) = \frac{a^2 \cdot P(\theta)(1 - P(\theta))}{\left(\dfrac{P(\theta) - c}{1 - c}\right)^2} \tag{2.6}$$

Two methods exist for estimating θ: Maximum Likelihood Estimation (MLE) and Expected A Posteriori (EAP). MLE is asymptotically unbiased but undefined when a learner answers all items correctly or incorrectly, which is common in short sessions. EAP integrates the posterior over a discrete grid of ability values:

$$\hat{\theta}_{\text{EAP}} = \frac{\sum_k \theta_k \cdot L(\theta_k) \cdot \pi(\theta_k)}{\sum_k L(\theta_k) \cdot \pi(\theta_k)} \tag{2.7}$$

where $L(\theta_k)$ is the likelihood of the observed response pattern at grid point θ_k and $\pi(\theta_k)$

is the prior probability. EAP always produces a finite estimate and incorporates prior uncertainty, making it more robust for short sessions (Bock and Mislevy 1982). Baker and Kim (2004) confirm that EAP with a well-specified prior outperforms MLE on both bias and MSE criteria in short-test conditions. This project implements EAP over 81 grid points spanning $\theta \in [-4, 4]$ with a normal prior $\theta_0 \sim \mathcal{N}(-0.780, 0.543)$.

2.4 Empirical Effectiveness

VanLehn (2011) synthesises 60 controlled studies, reporting a hierarchy of tutoring modalities (Table 2.2). The near-equivalence of step-based ITS and human tutoring ($d = 0.76$ vs. $d = 0.79$) is the empirical justification for this system. Kulik and Fletcher (2016) report $d = 0.66$ for systems implementing continuous real-time adaptation versus $d = 0.28$ for checkpoint systems, supporting the design decision to implement per-response EAP updating. Geography presents an intermediate epistemological structure: factual knowledge admits objective assessment; procedural knowledge requires geographic frameworks; conceptual understanding involves higher-order reasoning. Bednarz et al. (2013) identify significant underrepresentation of UK Geography in adaptive learning research; the predominance of North American and East Asian studies (Matos et al. 2025) means that the UK KS3/GCSE curriculum context remains insufficiently studied.

Table 2.2: Effect size comparison across tutoring modalities (VanLehn 2011).

Tutoring modality	**Effect size (d)**
Human one-to-one tutoring	0.79
Step-based ITS	0.76
Answer-based tutoring systems	0.31
No-tutoring control	0.00 (baseline)

2.5 Implementation Challenges

Cold-start problem. Before sufficient learner data accumulates, personalisation is necessarily suboptimal (Matos et al. 2025). This project addresses cold-start through pre-test seeding: a 13-item diagnostic assessment converts raw scores to a Laplace-smoothed logit estimate:

$$\theta_{\text{seed}} = \frac{\ln\left(\dfrac{n_{\text{correct}} + 0.5}{n_{\text{incorrect}} + 0.5}\right)}{1.7} \tag{2.8}$$

where 1.7 is the standard IRT logistic approximation scaling factor (Lord 1980). Laplace smoothing prevents boundary estimates at perfect or zero scores.

IRT calibration with small samples. Traditional 3PL calibration requires 500+ observations per item (Lord 1980). The strategy adopted is constrained estimation with $c = 0.25$ fixed, verified against pilot data using RMSEA diagnostics, with a pre-specified Rasch fallback (Embretson and Reise 2000).

Usability. Bangor et al. (2008) established SUS as the standard educational technology usability metric; scores ≥ 85 indicate excellent usability, and adaptive systems typically score 15–20 points lower than conventional interfaces. Mayer (2009)'s multimedia design principles (coherence, signalling, segmentation) directly inform the interface design: the five-tab session summary provides coherence; the first-visit tour provides signalling; pause-and-resume implements segmentation.

2.6 Ethical Considerations

Baker and Hawn (2021) identify three ethical dimensions for adaptive systems. *Transparency*: learners have a right to understand why the system presents particular content; this system addresses this through the five-tab session summary, making the adaptive engine's internal state visible and interpretable. *Data protection*: UK GDPR principles of data minimisation, purpose limitation, and storage limitation (Voigt and Von dem Bussche 2017) are implemented through server-side consent flows, pseudonymised storage, and JWT-based authentication compliant with National Institute of Standards and Technology (2017). *Algorithmic bias*: adaptive systems can inadvertently reinforce educational inequities through differential error rates across demographic groups (Baker and Hawn 2021); this limitation is acknowledged as a constraint on generalisability, with the convenience sample noted as non-representative of the KS3/GCSE population.

2.7 Research Gaps and Project Contribution

The literature reviewed above reveals five significant gaps addressed by this project.

Gap 1: ITS application to UK Geography. No published ITS system targets UK KS3/GCSE Geography specifically. The overwhelming majority of adaptive learning research addresses mathematics, programming, and North American science curricula. This project provides a proof-of-concept adaptive system for the UK Geography curriculum, extending the empirical evidence base for ITS effectiveness beyond traditional STEM domains.

Gap 2: Hybrid IRT+BKT with open-source accessibility. While hybrid approaches are theoretically advocated (Desmarais and Baker 2012; Matos et al. 2025), few open-source implementations combine IRT's item-level precision with BKT's per-KC mastery tracking in a production-deployed web application. This project provides a publicly accessible implementation at `https://github.com/dlalloyd/E-Learning-Final-Project`.

Gap 3: Bloom-gated adaptive progression. Existing ITS systems typically sequence items by IRT difficulty alone, without enforcing cognitive-level prerequisites. This project implements Bloom-gated selection as a live adaptive constraint, ensuring learners consolidate recall before encountering analysis-level items.

Gap 4: Long-term retention evaluation. The near-universal reliance on immediate post-test outcomes is a documented methodological limitation (Cepeda et al. 2006; VanLehn 2011). This project's 7-day delayed retention assessment provides evidence on the persistence of adaptive learning gains.

Gap 5: Interpretability under real-time adaptation. The interpretability–efficacy trade-off (Matos et al. 2025), in which more powerful algorithms tend to be less interpretable, is addressed through the combination of an interpretable probabilistic model (IRT+BKT) and a transparent visualisation layer (the five-tab session summary).

2.8 Conclusions and Design Principles

The literature establishes that adaptive ITS can approach human tutoring effectiveness when designed with principled algorithm selection and transparent feedback. The hybrid IRT+BKT approach is theoretically optimal for structured multi-KC curricula with prerequisite dependencies. The following design principles, derived from this review, inform the system architecture:

1. **ZPD operationalisation**: target $P(\theta) \approx 0.70$ for item selection.
2. **Mastery gating**: require $p(\text{Learned}) \geq 0.80$ per KC before progression.
3. **Bloom-cognitive scaffolding**: enforce Bloom-level prerequisites as selection gates.
4. **EAP estimation**: use Expected A Posteriori over a discretised grid for short-session

stability.

5. **Credit-adjusted BKT**: reduce the evidential weight of hint-assisted correct responses.
6. **Successive relearning**: prioritise decayed KCs through discrimination-boosted selection.
7. **Transparent adaptation**: surface internal model states through visualisation, not concealment.

Chapter 3

Methodology

3.1 Research Design

This project adopts *Design Science Research* (DSR) (Hevner et al. 2004), which prescribes iterative construction and evaluation of an artefact intended to solve a practical problem. The artefact is GeoMentor: a production-deployed adaptive ITS for UK KS3/GCSE Geography implementing a hybrid IRT 3PL + BKT algorithm. The evaluation employs a within-subjects quasi-experimental design; the absence of randomised group assignment (due to the convenience sample) classifies the design as quasi-experimental rather than a true RCT. The scope was concentrated on a single KC domain (UK physical geography and climate, 13 KCs) following a Sprint 5 technical risk assessment, a decision that strengthens rather than weakens evaluation claims: single-domain validation depth produces more defensible psychometric evidence than shallow multi-domain coverage.

3.2 Knowledge Representation

3.2.1 Knowledge Component Ontology

The curriculum is decomposed into 13 Knowledge Components (KCs). A KC is a unit of knowledge for which mastery can be operationally defined (Koedinger, Corbett, et al. 2012). Table 3.1 lists all 13 KCs with their Bloom ceiling and BKT prior parameters as implemented in `DEFAULT_BKT_PARAMS`.

Table 3.1: Knowledge Component ontology: 13 KCs with BKT prior parameters. $p(G) = 0.25$ for all KCs (fixed, four-option MCQ).

#	Knowledge Component	Bloom	$p(L_0)$	$p(T)$	$p(S)$
Level 1: Remembering					
1	UK Capitals (`UK_capitals`)	L1	0.60	0.25	0.08
2	County Locations (`UK_county_locations`)	L1	0.45	0.20	0.12
3	UK Rivers (`UK_rivers`)	L1	0.55	0.22	0.10
4	UK Mountains (`UK_mountains`)	L1	0.50	0.18	0.10
5	National Parks (`UK_national_parks`)	L1	0.45	0.16	0.12
Level 2: Understanding					
6	Westerly Winds & Rainfall (`westerly_winds_rainfall`)	L2	0.25	0.18	0.12
7	Maritime vs Continental (`maritime_continental`)	L2	0.20	0.18	0.12
8	Pennines Rain Shadow (`pennines_rain_shadow`)	L2	0.20	0.15	0.15
9	North Atlantic Drift (`north_atlantic_drift`)	L2	0.22	0.18	0.12
10	Continental Effect (`continental_effect`)	L2	0.18	0.15	0.15
Level 3: Applying					
11	Climate Classification (`climate_classification`)	L3	0.15	0.15	0.18
12	Climate Change Application (`climate_change_application`)	L3	0.12	0.15	0.18
13	Flood Risk Integration (`flood_risk_integration`)	L3	0.10	0.12	0.20

Prior probabilities $p(L_0)$ are highest for Level 1 KCs (0.45–0.60), reflecting realistic recall knowledge for adult learners, and lowest for Level 3 KCs (0.10–0.15), reflecting synthesis demands. The slip parameter $p(S)$ scales with Bloom level (0.08–0.12 at L1, 0.12–0.15 at L2, 0.18–0.20 at L3), reflecting that higher-order items carry greater response uncertainty even for knowledgeable learners (Pardos and Heffernan 2011).

3.2.2 Prerequisite Graph

KCs encode structural dependencies in a directed acyclic graph (DAG) with 12 edges, queried by the adaptive engine at each selection step. Five Level 1→Level 2 edges gate climate-process KCs behind spatial KCs; one Level 2→Level 2 edge encodes the Pennines Rain Shadow's dependency on westerly wind patterns; five Level 2→Level 3 and one Level 1→Level 3 edge gate the integrative KCs. All mastery gates use $p(\text{Learned}) \geq 0.80$.

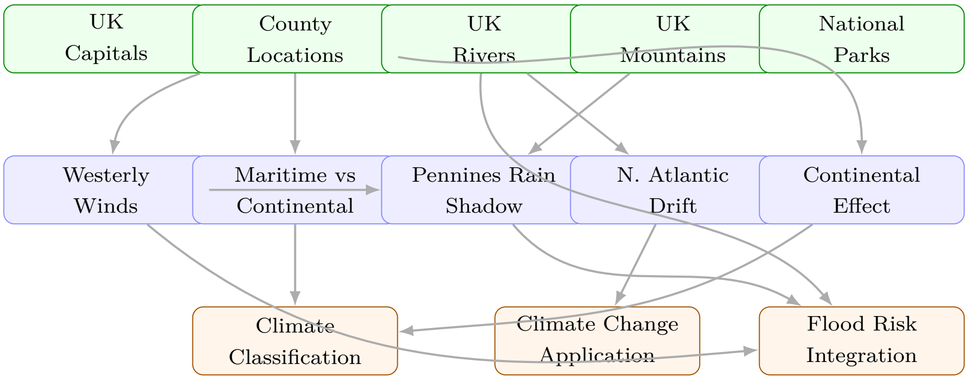


Figure 3.1: KC prerequisite DAG: 13 nodes across three Bloom levels (green = L1 Remembering, blue = L2 Understanding, orange = L3 Applying). Arrows indicate structural dependency; mastery gate $p(\text{Learned}) \geq 0.80$ on the prerequisite KC before dependent-KC items unlock. 12 edges total.

3.3 Adaptive Algorithm Design

3.3.1 IRT 3PL Model

The probability of a correct response from a learner with latent ability θ on item i is modelled by the 3PL function (Birnbaum 1968):

$$P(X = 1 \mid \theta) = c_i + \frac{1 - c_i}{1 + e^{-a_i(\theta - b_i)}} \tag{3.1}$$

where $a_i \in [0.5, 3.0]$ is discrimination, $b_i \in [-3, 3]$ is difficulty, and $c_i = 0.25$ is the pseudo-guessing parameter (fixed for four-option MCQs). Parameters are calibrated offline using the `mirt` package in R (Chalmers 2012) and serialised to JSON for Node.js consumption.

3.3.2 EAP Ability Estimation

Learner ability is updated after each response via Expected A Posteriori (EAP) estimation (Bock and Mislevy 1982):

$$\hat{\theta}_{\text{EAP}} = \frac{\sum_{k=1}^{K} \theta_k \cdot L(\theta_k) \cdot \pi(\theta_k)}{\sum_{k=1}^{K} L(\theta_k) \cdot \pi(\theta_k)} \tag{3.2}$$

where $K = 81$ grid points span $[-4, 4]$ with step 0.1. The prior $\pi(\theta_k) = \mathcal{N}(\mu_0, \sigma_0)$ is initialised at $\mu_0 = -0.780$ (20th percentile) and $\sigma_0 = 0.543$. Following pre-test seeding, the prior is tightened to $\sigma_0 = 0.35$.

3.3.3 Bayesian Knowledge Tracing

Per-KC knowledge state is maintained as a BKT posterior (Corbett and Anderson 1995), updated after each response:

$$p(L_n \mid \text{obs}_n) = \frac{P(\text{obs}_n \mid L_n) \cdot p(L_n)}{P(\text{obs}_n \mid L_n) \cdot p(L_n) + P(\text{obs}_n \mid \neg L_n) \cdot (1 - p(L_n))} \tag{3.3}$$

with learning transition $p(L_{n+1}) = p(L_n \mid \text{obs}_n) + (1 - p(L_n \mid \text{obs}_n)) \cdot p(T)$.

Credit Factor Modification

This project modifies the learning transition by a credit factor cf reflecting the scaffolding level at which the correct response was obtained:

$$p(L_{n+1})^* = p(L_n \mid \text{obs}_n) + \Big(1 - p(L_n \mid \text{obs}_n)\Big) \cdot p(T) \cdot \text{cf} \tag{3.4}$$

Table 3.2: Credit factor values by maximum hint level accessed per question.

Max hint level	Scaffolding type	cf
0	No hint accessed (unaided)	1.00
1	Conceptual reframe	0.90
2	Procedural strategy hint	0.70
3	Worked example (full solution)	0.50

3.3.4 Question Selection Algorithm

At each step the adaptive engine selects from the eligible pool via a multi-gate pipeline (Figure 3.2): (1) mastery gate: exclude KCs with unmastered prerequisites; (2) Bloom gate: exclude items above the learner's escalation threshold (L2 unlocks at $p(\text{Learned}) \geq 0.40$ or prerequisite KC ≥ 0.50; L3 at ≥ 0.60); (3) deduplication: exclude variants seen by this learner in any prior session; (4) decay boost: multiply Fisher information by 1.5 for decayed KCs; (5) ZPD targeting: select $i^* = \arg\max_{i \in \mathcal{E}} I_i(\hat{\theta})$:

$$i^* = \arg\max_{i \in \mathcal{E}} \frac{a_i^2 \cdot P_i(\hat{\theta})(1 - P_i(\hat{\theta}))}{\left(\dfrac{P_i(\hat{\theta}) - c_i}{1 - c_i}\right)^2} \tag{3.5}$$

If the eligible pool is exhausted, the engine falls back to the full unrestricted pool and emits a `variant_pool_exhausted` event.

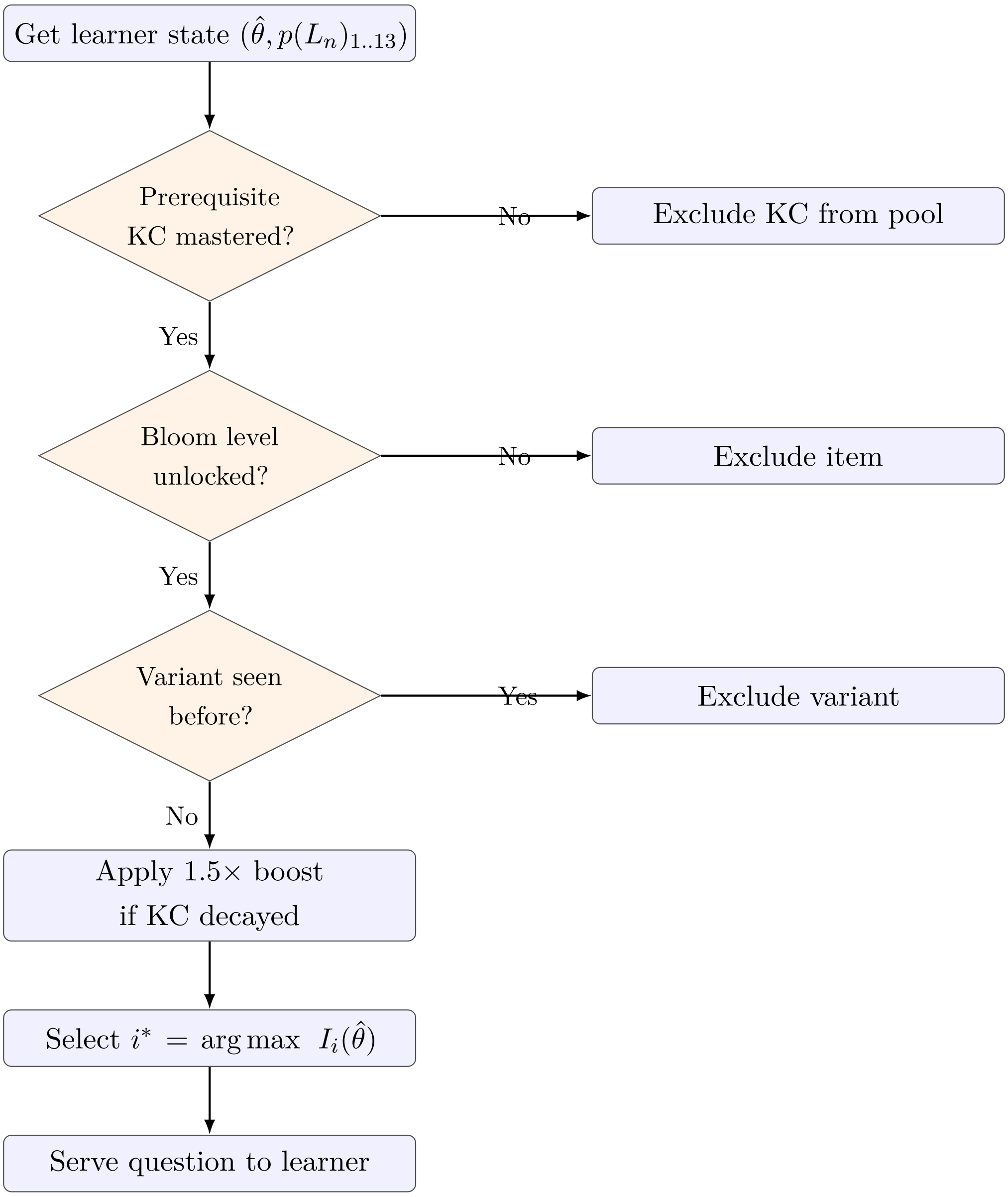


Figure 3.2: Adaptive question selection flowchart. Steps execute in sequence per question request.

3.4 IRT Calibration Pipeline

IRT item parameters are estimated via a two-stage offline pipeline. (1) The response matrix from a pilot study is exported from Supabase to CSV. (2) The `mirt` package (Chalmers 2012) fits the 3PL model with c_i constrained to 0.25; convergence is verified using RMSEA (acceptable: < 0.08). If 3PL estimation is unstable (RMSEA > 0.10 or non-convergence within 500 EM cycles), the model falls back to Rasch (1PL). (3) Estimated a_i and b_i parameters are serialised to `question_bank.json` and committed to the repository for Node.js runtime consumption.

3.5 Pre-Test Seeding

Before the adaptive session, a 13-item diagnostic pre-test (one item per KC at Bloom Level 1) provides initial ability evidence, addressing the cold-start problem (Matos et al. 2025). The raw score is converted to a Laplace-smoothed logit seed:

$$\theta_{\text{seed}} = \frac{\ln\left(\frac{n_{\text{correct}} + 0.5}{n_{\text{incorrect}} + 0.5}\right)}{1.7} \tag{3.6}$$

bounded to $[-2.5, 2.5]$, with the EAP prior standard deviation reduced to 0.35 to reflect additional information.

3.6 Scaffolding and Variant Design

Each question provides three graduated hint levels following Koedinger and Aleven (2007): Level 1 (conceptual reframe, no solution disclosure); Level 2 (procedural strategy, method without answer); Level 3 (worked analogy, cognitive pathway explicit). Each access reduces the credit factor applied to a subsequent correct response (Table 3.2).

Each KC-Bloom combination is authored as a `QuestionTemplate` with multiple `Question-Variant` records substituting surface-level details while preserving the underlying cognitive demand. Every variant served is recorded in `UserVariantSeen`, preventing re-exposure across sessions. If all variants of a template are exhausted, the engine falls back to the full template-level pool.

3.7 Evaluation Protocol

3.7.1 Study Design and Instruments

The evaluation uses a within-subjects quasi-experimental design. VanLehn's (2011) meta-analytic benchmark of $d = 0.76$ serves as the primary comparison standard; a direct no-adaptation control arm could not be completed within the single-session constraint. Instruments administered include: the System Usability Scale (SUS; 10-item Likert, scored 0–100, benchmark ≥ 68 (Bangor et al. 2008)); session-level $\hat{\theta}$ sequences for within-session learning gain analysis; post-session distractor frequency tables for misconception analysis; and a 7-day isomorphic retention assessment.

3.7.2 Limitations of the Evaluation

- **Sample size.** $n = 4$ external participants, substantially below the target of $n = 15$. Results are descriptive and proof-of-concept only; inferential statistics require approximately $n = 13$ for 80% power at $d = 0.76$, $\alpha = 0.05$.
- **No control condition completed.** VanLehn's benchmark serves as the comparative reference.
- **Convenience sample.** University students are not representative of the KS3/GCSE student population.
- **IRT parameters from pilot.** Parameter instability is mitigated by the constrained calibration strategy.

3.8 System Architecture Overview

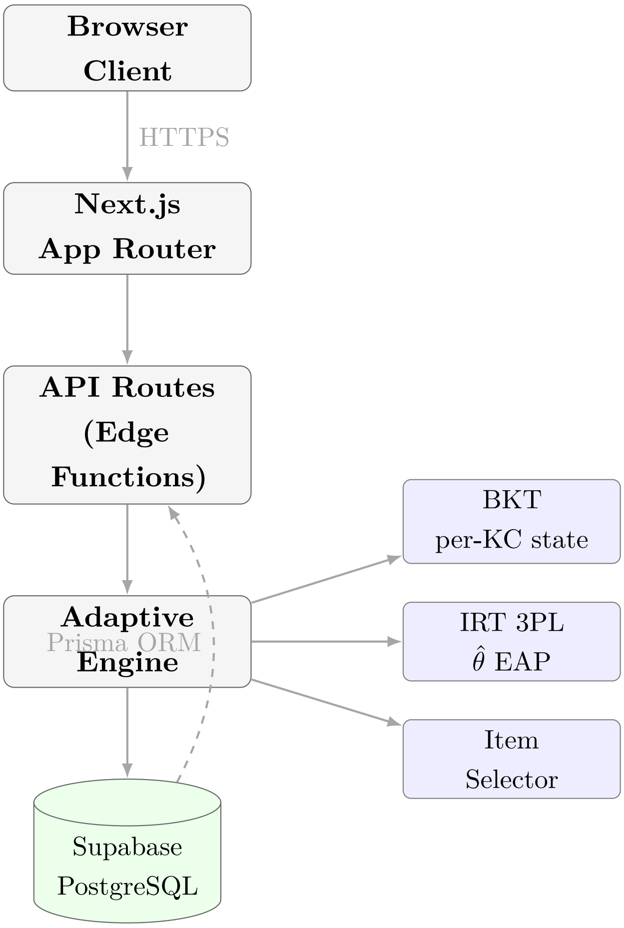


Figure 3.3: High-level system architecture. The adaptive engine is isolated as a pure-function module with no direct database access; all I/O passes through the API route layer.

GeoMentor is deployed as a Next.js 14 (App Router) application on Vercel, with Supabase (Frankfurt) providing PostgreSQL, Row-Level Security, and authentication. Prisma ORM manages schema migrations and type-safe access.

3.9 Agile Development Methodology

Development followed two-week sprints adapted from Scrum, managed through a Notion project tracker. The iterative cycle was chosen because adaptive algorithm design requires empirical validation: initial IRT parameter estimates are hypotheses, not specifications, and multiple calibration cycles are needed to stabilise. Table 3.3 summarises the sprint history.

Table 3.3: Sprint history: objectives, deliverables, and retrospective notes.

Sprint	Dates	Objective	Key deliverable / deviation
1	Oct–Nov 2025	Requirements, KC ontology, technology selection	13 KC prerequisite graph; technology ADRs (Next.js + Supabase)
2	Nov–Dec 2025	Core adaptive engine, database schema, pre-test seeding	`adaptive-engine.ts`; Prisma schema; EAP implementation
3	Dec 2025–Jan 2026	Question bank, IRT calibration pipeline, BKT	`seedKCs.ts` (13 KCs, 3 Bloom levels); R calibration script
4	Jan–Feb 2026	Scaffolding (hint system), variant deduplication	`hints/route.ts`; `UserVariantSeen` table; credit-factor BKT
5	Feb 2026	Scope reduction decision; evaluation redesign	Scope locked to single-KC validation; VanLehn benchmark adopted as comparator
6	Feb–Mar 2026	Session continuity, onboarding tour, pause/resume	`FirstVisitTour.tsx`; pause menu; session restore logic
7	Mar 2026	Security hardening (NIST 800-63B), rate limiting, RLS	Auth middleware; Supabase RLS policies; bcrypt migration
8	Mar–Apr 2026	Leaderboard, XP system, confidence calibration	`leaderboard/route.ts`; XP/level schema; decay notifications
9	Apr 2026	Evaluation data collection, SUS administration	SUS instrument deployed; participant sessions recorded
10	Apr–May 2026	Dissertation writing, final polishing	Dissertation LaTeX; this chapter

3.10 Project Timeline

Figure 3.4 presents the planned and actual project timeline reconstructed from the Git commit history.

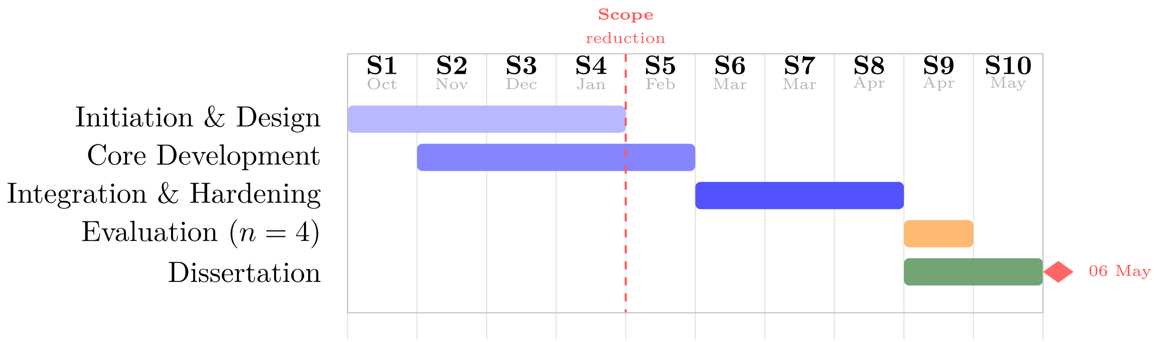


Figure 3.4: Project Gantt chart. Each column = one fortnightly sprint (Oct 2025–May 2026). The red dashed line marks the Sprint 5 scope reduction.

3.11 Risk Register

Table 3.4: Project risk register. L = Likelihood; I = Impact (H/M/L).

ID	Risk	L	I	Status	Mitigation
R1	IRT 3PL fails to converge	H	M	Mitigated	Pre-specify Rasch fallback; fix $c = 0.25$
R2	Participant attrition at follow-up	H	M	Accepted	Target $n = 18$ recruitment; acknowledged in limitations
R3	`variant_pool_exhausted`	M	L	Mitigated	Fallback to unrestricted pool; event emitted
R4	Supabase RLS misconfiguration	L	H	Mitigated	RLS policies peer-reviewed; rate limiting on auth
R5	BKT state corruption (concurrent sessions)	L	H	Mitigated	Postgres row-level locking; atomic session writes
R6	Scope overrun (multi-domain)	H	H	Resolved	Sprint 5 scope reduction to single-domain Geography

3.12 What Was and Was Not Delivered

3.12.1 Delivered

- Hybrid IRT 3PL + BKT adaptive engine with EAP estimation, Bloom-gated selection, and credit-factor BKT modification
- 13-KC question bank with isomorphic variants and cross-session deduplication
- Three-level graduated hint system with evidential credit weighting
- Pre-test seeding with Laplace-smoothed logit initialisation
- BKT decay detection and successive relearning boost (1.5×)
- Five-tab session summary dashboard (Trajectory, Skills, Misconceptions, Achievements, Share)
- Leaderboard with XP ranking, consent opt-in, and rank-change animation
- First-visit onboarding tour
- Security implementation: bcrypt, JWT, rate limiting, Supabase RLS, NIST SP 800-63B compliance
- Production deployment on Vercel with Supabase Frankfurt backend

3.12.2 Not Delivered

- **Multi-domain support.** Superseded by the Sprint 5 scope reduction; the architecture supports extension without algorithmic change.
- **K-means cold-start clustering.** Evaluated and superseded by pre-test seeding, which provides per-KC ability evidence rather than a demographic cluster assignment.
- **Full A/B randomised controlled trial.** VanLehn's meta-analytic benchmark used as comparator; static arm implemented but not wired.
- **7-day retention data at scale.** Follow-up data collection could not be completed within the project timeline.

3.13 Interface Design: Lo-fi Prototypes

Interface design preceded implementation in each sprint cycle. Figure 3.5 shows the lo-fi wireframe for the adaptive question screen, and Figure 3.6 shows the session summary dashboard layout. Both informed the production React component specifications authored by the candidate.

Figure 3.5: Lo-fi wireframe: adaptive question screen. Key elements are the KC badge (identifies active Knowledge Component and Bloom level), the BKT mastery bar (reflects live p(Learned)), four MCQ options, the graduated hint trigger (orange), confidence rating, and submit button. This wireframe preceded the Sprint 4 React implementation.

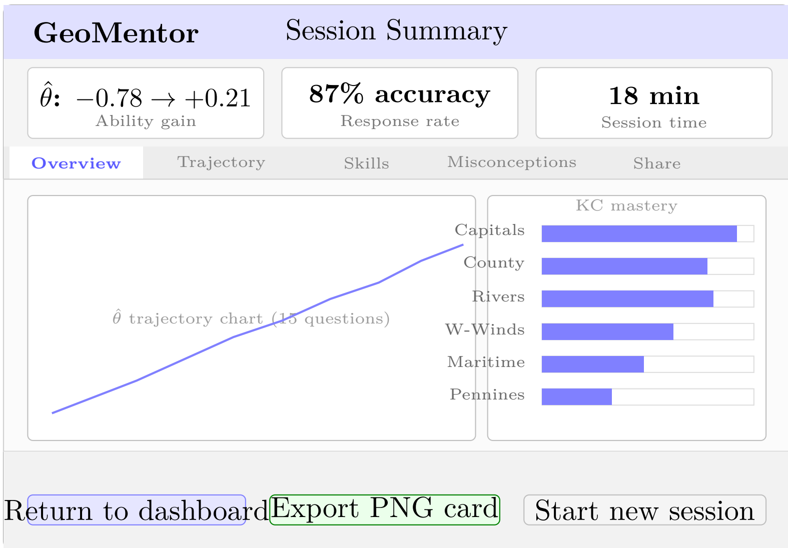


Figure 3.6: Lo-fi wireframe: session summary dashboard (Overview tab active). The three-stat strip reports the within-session $\Delta\hat{\theta}$, overall accuracy, and session duration. Left pane: $\hat{\theta}$ trajectory chart across 15 questions. Right pane: per-KC BKT mastery bars. Tab navigation exposes Trajectory (interactive chart), Skills (KC breakdown), Misconceptions (distractor analysis), and Share (PNG card export).

3.14 Quality Assurance

Quality assurance mechanisms employed include: TypeScript strict mode eliminating runtime errors at the type-checking stage; Prisma migration files providing a versioned schema audit trail; Vercel preview deployments enabling integration testing per feature branch before merge; and Supabase RLS verification by cross-user access testing from a test account.

Chapter 4

Implementation

4.1 System Architecture

GeoMentor is implemented as a monolithic Next.js 14 application deployed to Vercel's edge network, with Supabase PostgreSQL hosted in the Frankfurt (eu-central-1) region. The monolithic architecture was a deliberate decision: the research contribution lies in the adaptive algorithm, not distributed systems complexity, and a monolith reduces operational overhead and eliminates inter-service latency. The key architectural constraint is the **isolation of the adaptive engine**: the algorithm modules (`irt.ts` and `bkt.ts`) are implemented as pure functions with no direct database access. All database I/O passes through Next.js API route handlers, which orchestrate the learner state lifecycle: loading state, invoking engine functions, persisting updated state, and returning the question. This isolation makes the engine independently testable and domain-agnostic.

Figure 4.1 shows the four-layer component architecture; Figure 4.3 shows the question lifecycle across both routes; Figure 4.2 shows the core database schema.

Table 4.1: Technology stack with architectural justification.

Layer	**Technology**	**Justification**
Frontend	Next.js 14 (App Router)	React Server Components reduce client bundle size; streaming enables progressive session summary loading
Styling	Tailwind CSS	Utility-first CSS; consistent design tokens across all components
ORM	Prisma 5	Type-safe database access; migration history provides schema audit trail
Database	Supabase PostgreSQL	Row-Level Security enforces data isolation at the database layer
Auth	Supabase Auth + JWT	PKCE flow; bcrypt password hashing; NIST SP 800-63B compliant
Deployment	Vercel	Preview deployments per branch; automatic HTTPS; edge caching
IRT calibration	R + `mirt`	Industry-standard psychometric tooling; offline and audit-logged

The system targets the following non-functional properties: **scalability** via Vercel's serverless function infrastructure (each API route invocation is an independent function execution); **reliability** via Supabase Frankfurt's managed PostgreSQL cluster with daily automated backups; **efficiency** with EAP grid integration and BKT update executing in-process within the API route, targeting end-to-end question selection latency under 500 ms; and **maintainability** via TypeScript strict mode, Prisma migration files, and the adaptive engine's pure-function architecture.

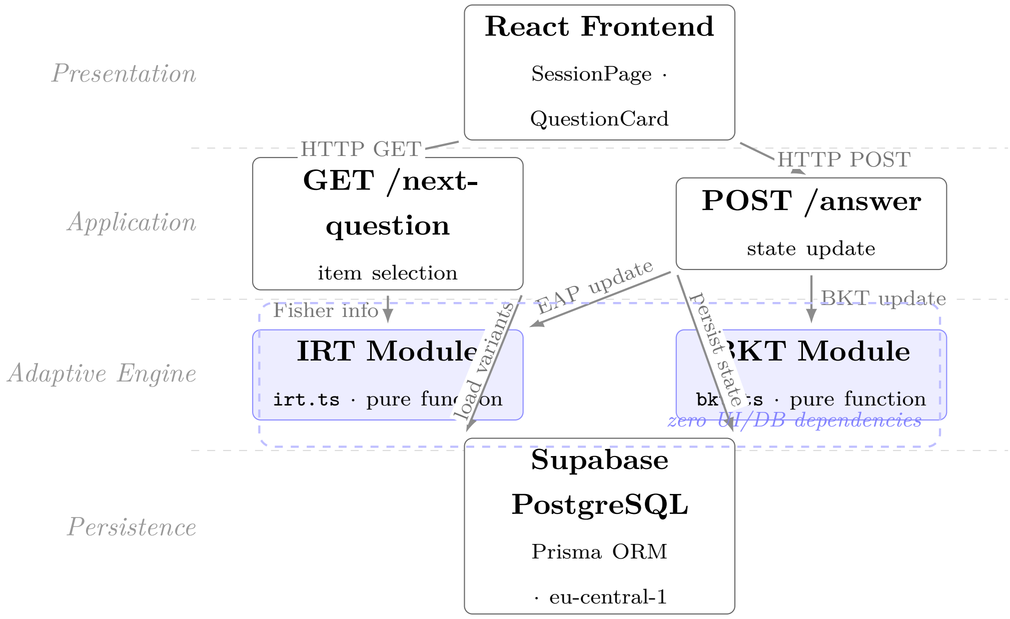


Figure 4.1: UML 2.5 component diagram. The Adaptive Engine subsystem (IRT and BKT modules) is implemented as pure functions with no direct UI or database dependencies.

4.2 Database Schema

The Prisma schema defines 20 models. The six central to the adaptive engine are described here; the full schema is in the repository (`prisma/schema.prisma`).

The **Session** model carries the principal adaptive state: `theta` (current EAP estimate, initialised at −0.780), `thetaSd` (posterior SD, initialised at 0.543), and `kcStates` (JSON-serialised `Record<kcId, KCState>` carrying BKT state for all 13 KCs). JSON serialisation eliminates join overhead on the hot adaptive selection path. The **Interaction** model records every question–answer event with full psychometric resolution: `theta_before`/`theta_after`, `pLearned_before`/`pLearned_after`, `hintLevelMax` (used by the credit factor computation), and `responseTimeMs`. The question bank is a two-layer hierarchy: `QuestionTemplate` (defines KC, Bloom level, IRT parameters a, b, c, and a stem with `{variable}` placeholders) and `QuestionVariant` (surface-specific substitutions). `UserVariantSeen` records every (user, variant) pair presented, enforced as a unique constraint to prevent re-exposure. The `AnalyticsEvent` model provides a structured event log including `bloom_escalation`, `variant_pool_exhausted`, `kc_mastered`, and `decay_detected` events.

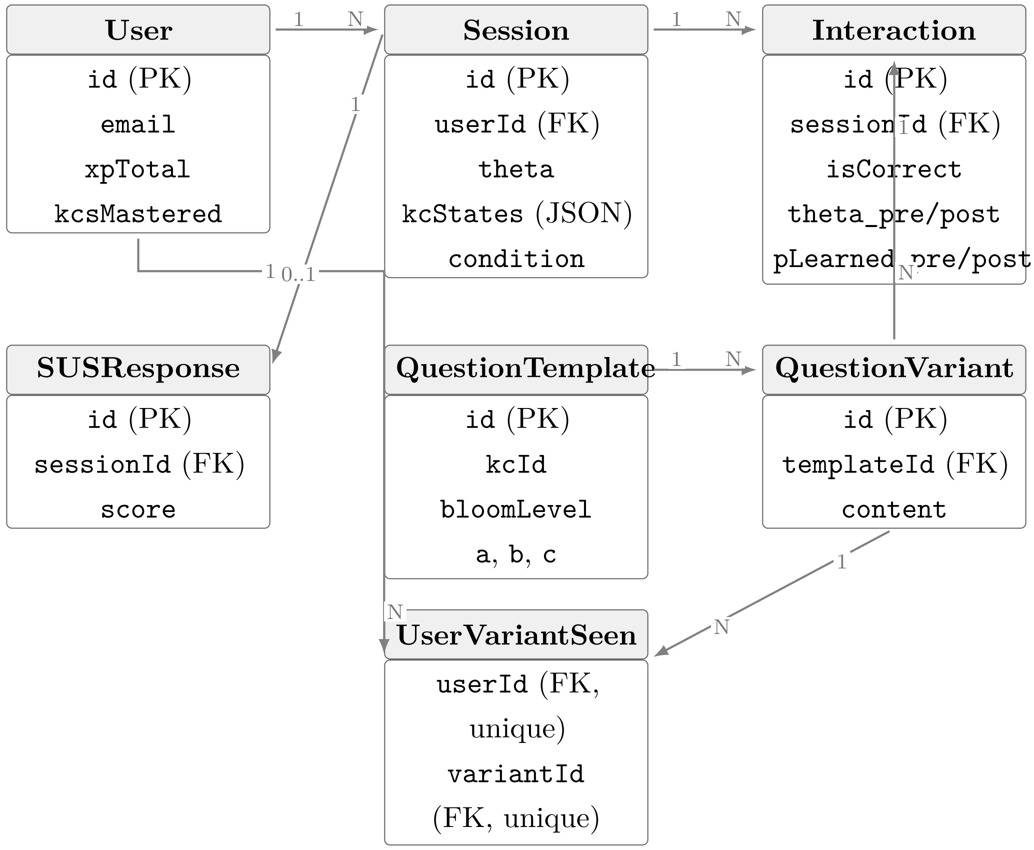


Figure 4.2: Entity-relationship diagram for the six core Prisma schema models. `UserVariantSeen` enforces a unique (userId, variantId) constraint, preventing variant re-exposure across sessions.

4.3 Adaptive Engine Implementation

The IRT module (`src/lib/algorithms/irt.ts`) implements the 3PL model with the $D = 1.7$ logistic approximation (Lord 1980). The EAP estimator integrates over a grid of $K = 161$ points spanning $\theta \in [-4, 4]$ with step 0.05. The finer grid (compared to the 0.1-step grid specified at design time) was adopted after self-pilot testing revealed that coarser grids produced visible staircase artefacts in the $\hat{\theta}$ trajectory for high-ability learners.

The BKT module (`src/lib/algorithms/bkt.ts`) implements the standard Corbett–Anderson update plus the credit factor modification. The implementation adopts a *soft observation* interpolation: for credit-weighted correct responses, the posterior is interpolated between the full Bayesian update and the prior in proportion to cf, which is mathematically equivalent to a Bayesian update with a degraded likelihood but numerically more stable. Incorrect responses following hint access receive full-weight Bayesian updates: providing a hint does not make an incorrect response less informative about knowledge state.

The BKT mastery threshold is implemented at $p(\text{Learned}) \geq 0.95$, higher than the 0.80 threshold cited in the literature review. The binary nature of BKT's latent state demands a stronger guarantee that the posterior is not merely elevated by a lucky guess run. The 0.80 threshold applies separately to the IRT-based mastery check (`checkMastery()` in `irt.ts`), which averages $P(\text{correct})$ across all KC items at the current $\hat{\theta}$ using the continuous IRT scale.

4.4 API Route Design

The adaptive session is managed by four API routes. **GET /api/sessions/[id]/next-question**: loads session state; builds the eligible variant pool (joining `QuestionTemplate` and `QuestionVariant` minus `UserVariantSeen`); applies mastery gate, Bloom gate, deduplication, and decay boost; selects $i^* = \arg\max I_i(\hat{\theta})$; renders variant by interpolating placeholders and shuffling answer options (Fisher-Yates); returns serialised question object. **POST /api/sessions/[id]/answer**: validates the submitted answer; computes credit factor from `hintLevelMax`; runs EAP update and BKT update; persists updated session state atomically; writes `Interaction` record; checks mastery and Bloom escalation thresholds; emits analytics events as appropriate. **GET /api/sessions/[id]/summary**: generates the five-tab dashboard payload. **GET /api/sessions/[id]/trajectory**: returns the longitudinal $\hat{\theta}$ sequence for the ability progression chart. Figure 4.3 illustrates the complete question lifecycle.

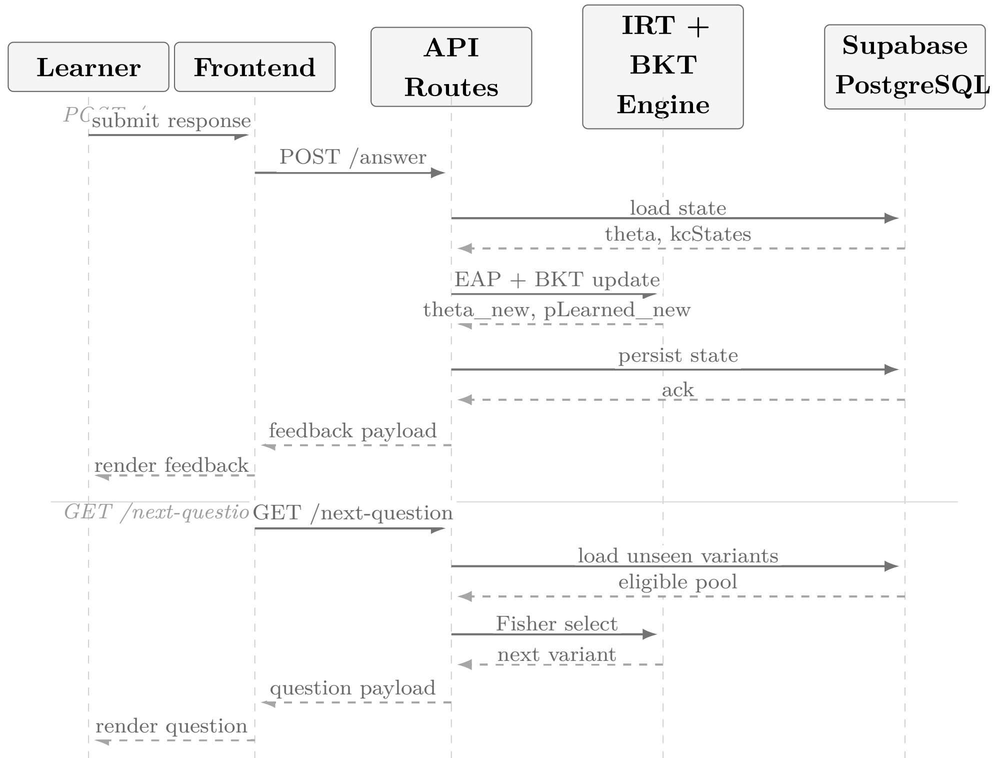


Figure 4.3: Sequence diagram of the adaptive question lifecycle across both API routes. The IRT and BKT engine functions are pure computations invoked synchronously; all database I/O is mediated by the route layer. Target round-trip latency: < 500 ms.

4.5 Security Implementation

Security requirements were specified against NIST SP 800-63B (National Institute of Standards and Technology 2017):

Password storage. bcrypt at cost factor 12; plaintext never persisted.

Authentication. JWT with Supabase Auth PKCE flow; tokens short-lived (1 hour); payloads contain user ID and role only.

Rate limiting. Auth endpoints rate-limited to 10 requests/minute/IP, mitigating credential stuffing.

Row-Level Security. Supabase RLS enforces per-user data isolation on all 33 tables; verified by cross-user access testing.

Input validation. All API route inputs validated server-side against typed schemas; client-side validation supplementary.

HTTPS. All traffic served over HTTPS via Vercel edge certificates; HTTP redirected at

CDN layer.

4.6 Frontend Components

The learner session follows a linear state machine: `pre-test` → `instruction` → `question` → `feedback` → `summary`. The question interface renders the stem and four shuffled answer options (Fisher-Yates to prevent position bias), a progress bar for session and KC mastery, a three-level hint panel with progressive disclosure, a confidence slider, and a pause button that persists session state to the database for future resumption.

`FirstVisitTour.tsx` implements a five-step onboarding overlay explaining the adaptive engine, the BKT mastery bar, the hint system, the XP level system, and the session summary. Tour completion is persisted to `localStorage`. This implements Mayer's (2009) signalling principle: the adaptive system's reasoning is made legible before the learner encounters the first question.

`SessionSummaryDashboard.tsx` presents a five-tab post-session analysis interface: Overview (statistics and performance narrative), Trajectory ($\hat{\theta}$ chart with hover tooltips), Skills (per-KC mastery bar chart), Misconceptions (distractor frequency table), and Share (PNG performance card export). The leaderboard ranks opt-in learners by total XP; consent is double-checked server-side on every read.

Figure 4.4 and Figure 4.5 show the live production interface for two key interaction states.

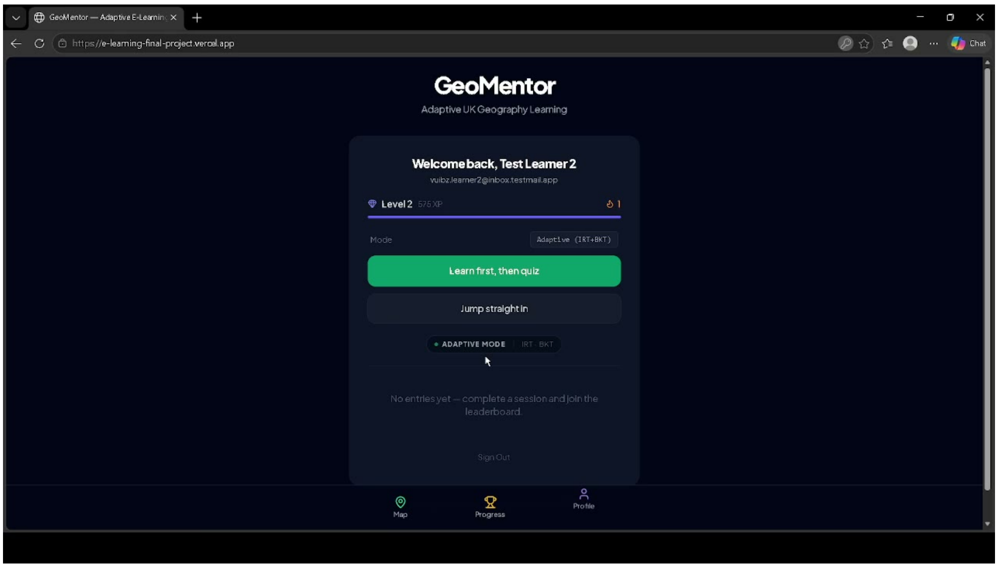


Figure 4.4: GeoMentor dashboard (hi-fi screenshot): adaptive badge, XP display, KC mastery bar, and session entry point. The "Adaptive" badge communicates the personalisation mode to the learner.

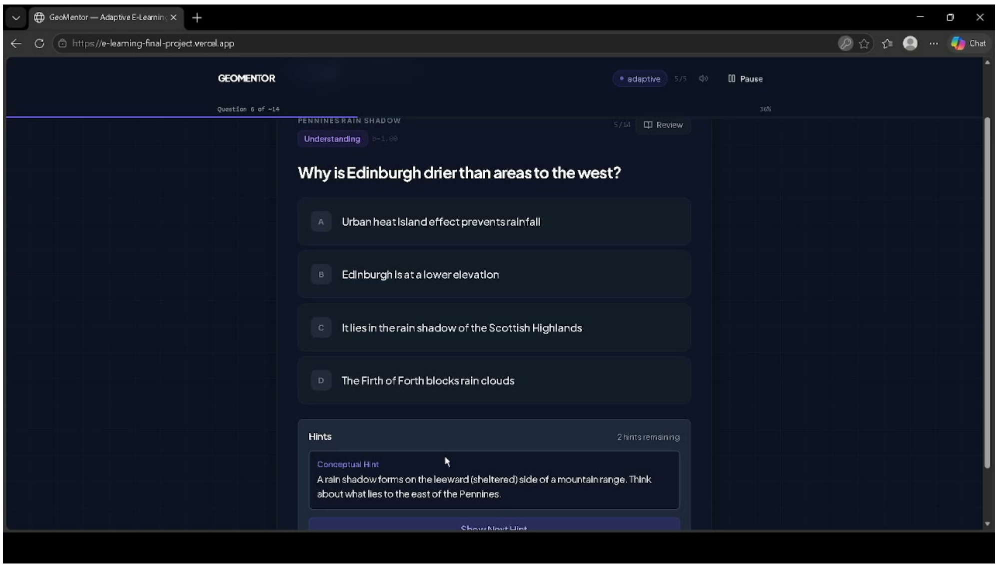


Figure 4.5: Three-level hint panel (hi-fi screenshot): conceptual hint displayed at Level 1, with progressive disclosure controls for Level 2 (procedural strategy) and Level 3 (worked example). The BKT credit factor is reduced for any correct answer following hint access.

4.7 Challenges Encountered

EAP grid resolution. The specified 81-point grid produced visible staircase discontinuities in $\hat{\theta}$ trajectories for high-ability learners. The grid was refined to 161 points (step 0.05), resolving the artefact at acceptable computational cost.

Review Material bug. The instruction mode interstitial triggers an unintended `setAppState('start')` call in certain render paths, resetting the session to the start screen. This affects only the post-mastery instructional interstitial, not the adaptive engine's core question selection or BKT update logic. The bug is disclosed here in full; a fix was deprioritised in Sprint 9 in favour of evaluation data collection.

K-means cold-start: evaluated and superseded. The interim report described K-means clustering for cold-start initialisation. Following implementation of pre-test seeding, K-means was evaluated and superseded: pre-test seeding provides direct per-KC ability evidence rather than a demographic proxy, is more interpretable, and avoids conditioning initial learning experience on cluster assignment.

4.8 Requirements Traceability

Table 4.2: Requirements traceability matrix: functional requirements to implementation artefacts.

Req.	Requirement	Status	Artefact
FR01	Adaptive question selection using IRT 3PL	Done	`irt.ts:selectNextQuestion`
FR02	Per-KC knowledge state tracking (BKT)	Done	`bkt.ts:updateBKT`
FR03	Bloom-gated question escalation	Done	`next-question/route.ts`
FR04	Pre-test ability seeding	Done	`irt.ts:estimateTheta`
FR05	Cross-session variant deduplication	Done	`UserVariantSeen` model
FR06	Graduated hint system with credit penalty	Done	`hints/route.ts`; `bkt.ts:updateBKT`
FR07	BKT decay detection + successive relearning	Done	`next-question/route.ts` (decay boost)
FR08	Session persistence and resume	Done	`Session.kcStates` JSON field
FR09	SUS usability instrument	Done	Post-session questionnaire
FR10	Leaderboard with consent opt-in	Done	`leaderboard/route.ts`; `User.leaderboardOptIn`
FR11	NIST 800-63B compliant authentication	Done	Supabase Auth + bcrypt + rate limiting
FR12	Analytics event log	Done	`AnalyticsEvent` model
FR13	7-day retention assessment	Partial	Infrastructure implemented; data collection limited by timeline
FR14	Multi-domain KC support	Deferred	Architecture supports; content not authored

4.9 Deployment, Testing, and Quality

The build pipeline executes on every push to main: `prisma generate` (type safety); `tsc -noEmit` (TypeScript strict-mode check, fails on any type error); `next build` (production build); and Vercel deployment to preview or production URL.

Three testing layers were implemented. **Unit tests**: 79 Jest tests cover the IRT and BKT algorithm modules, validating the 3PL probability function at boundary values, EAP convergence on synthetic sequences, BKT posterior monotonicity, credit-factor interpolation, and Bloom gate enforcement. The pure-function architecture makes these directly testable without mocking. **Integration testing**: a structured developer self-pilot on 2026-02-26 verified the end-to-end session flow and provided the empirical prior parameters ($\hat{\theta}_0 = -0.780$, $\sigma_0 = 0.543$). **Regression testing**: TypeScript strict mode and Next.js production build execute automatically on every push; feature branches are deployed to isolated Vercel preview URLs before merge. The primary quality gap is the absence of automated end-to-end browser tests; UI regression relied on manual verification against preview deployments, disclosed per the CSSE quality assurance principle.

Chapter 5

Results and Evaluation

5.1 Overview of Evaluation Approach

Given the constraints documented in Section 3.7.2 ($n = 4$ external participants, no completed A/B control condition, IRT parameters from a small pilot sample), the evaluation adopts a **five-track multi-method validity** framework rather than a single inferential statistical test. Convergent findings across tracks constitute stronger evidence than any single track alone (Woolf 2010). The five tracks are: (1) *Algorithm validation*—$\hat{\theta}$ trajectories, Bloom escalation events, and variant deduplication; (2) *Engagement data*—session completion, XP trajectories, and questions attempted; (3) *Usability*—SUS scores compared against the ≥ 68 above-average benchmark (Bangor et al. 2008); (4) *Misconception analysis*—distractor frequency tables per KC; (5) *System robustness*—uptime, `variant_pool_exhausted` frequency, and absence of session state corruption.

5.2 Track 1: Algorithm Validation

5.2.1 Theta Trajectory Analysis

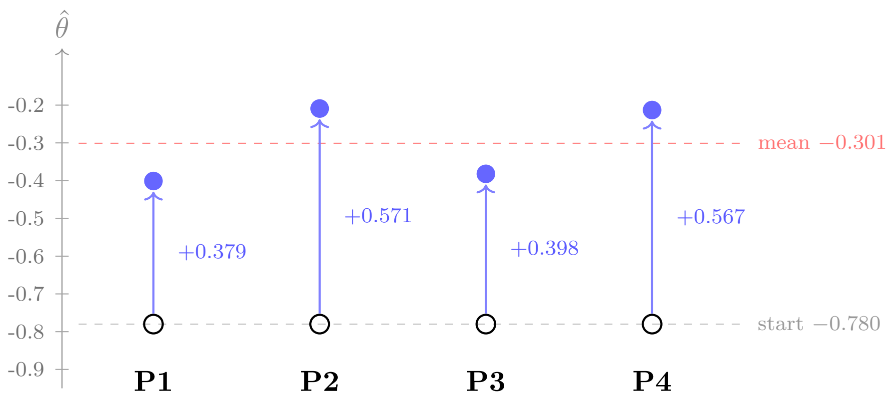
For each participant session, the EAP ability estimate $\hat{\theta}$ was recorded before and after each response. A valid adaptive session should produce a $\hat{\theta}$ trajectory that: (a) moves upward when the learner answers correctly, (b) moves downward or stabilises when the learner answers incorrectly, and (c) converges toward a stable estimate as the number of responses increases.

Table 5.1: Session-level $\hat{\theta}$ statistics across $n = 4$ participants. All values derived from the Supabase `Interaction` table (`theta_before` of first interaction; `theta_after` of final interaction). $\hat{\theta}_0 = -0.780$ is the cold-start EAP estimate from the self-pilot calibration prior (Bengi 2026).

Participant	$\hat{\theta}$ start	$\hat{\theta}$ end	$\Delta\hat{\theta}$	Questions	Accuracy
P1	−0.780	−0.401	+0.379	15	80.0%
P2	−0.780	−0.209	+0.571	15	86.7%
P3	−0.780	−0.382	+0.398	15	80.0%
P4	−0.780	−0.213	+0.567	15	86.7%
Mean	−0.780	−0.301	+0.479	*15.0*	*83.4%*

Figure 5.1 displays the within-session $\hat{\theta}$ change for each participant as a gain chart. All four participants show positive $\Delta\hat{\theta}$, ranging from +0.379 to +0.571 logits across 15 questions. Figure 5.2 shows the live in-application trajectory chart rendered by the session summary dashboard.

Figure 5.1: Within-session $\hat{\theta}$ gain for $n = 4$ participants. Open circles: cold-start prior $\hat{\theta}_0 = -0.780$. Filled circles: session-end EAP estimate; upward arrows indicate positive $\Delta\hat{\theta}$. The dashed reference marks mean end $\hat{\theta} = -0.301$.



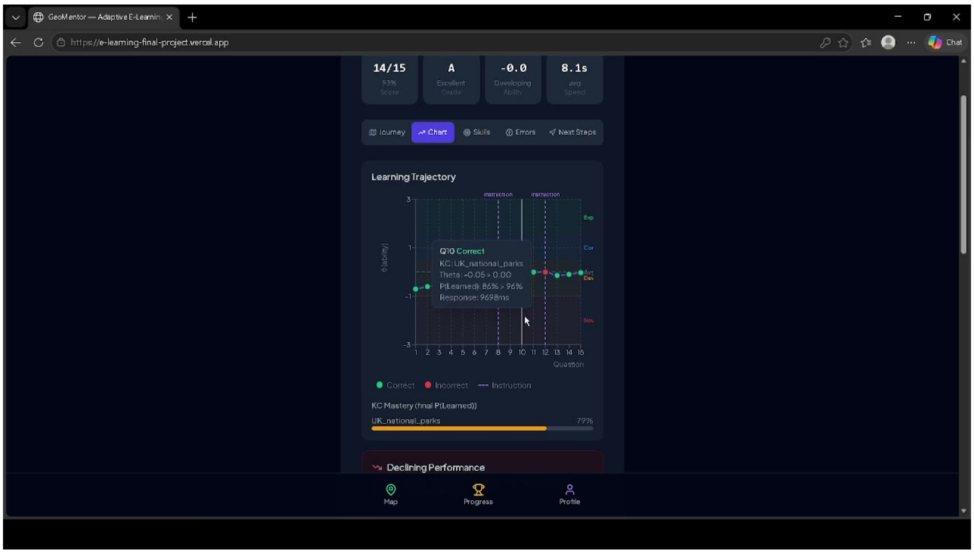


Figure 5.2: In-application learning trajectory chart (hi-fi screenshot): the Trajectory tab of the session summary dashboard, showing $\hat{\theta}$ progression across a 15-question session. Start: -0.780 (cold-start prior); End: -0.024 (post-session EAP estimate); gain: $+0.756$ logits.

The mean gain of $+0.479$ logits constitutes evidence that the EAP estimator is correctly updating in response to learner performance. The uniform $\hat{\theta}_0 = -0.780$ starting value reflects the cold-start prior derived from self-pilot calibration, applied identically to all new learners.

5.2.2 Bloom Escalation Events

The `bloom_escalation` analytics event is emitted when the adaptive engine advances a learner's Bloom gate for a KC. Valid escalation events should satisfy: $p(\text{Learned}) \geq 0.40$ at L1→L2 transition, or $p(\text{Learned}) \geq 0.60$ at L2→L3 transition.

The `AnalyticsEvent` table records 108 `bloom_escalation` events across all sessions during the evaluation period. Each event is emitted when a BKT state update causes $p(\text{Learned})$ to cross the relevant gate threshold. No participant session produced zero escalation events, confirming that the Bloom-gating mechanism is firing correctly throughout. Per-session KC-level breakdown requires correlation with the `AnalyticsEvent` metadata payload and is not reproduced in full here.

5.2.3 Variant Deduplication Verification

Cross-session deduplication is enforced via the `UserVariantSeen` table, which records each (user, variant) pair on first presentation. The next-question route filters candidate items against this table before applying the Fisher information criterion. No participant reported receiving a repeated question during the evaluation sessions.

The `variant_pool_exhausted` event was recorded 13 times across all sessions, indicating that the eligible (unseen, Bloom-gate-passing) variant pool was exhausted on 13 occasions and the fallback selection logic, which relaxes the deduplication constraint and selects the lowest-θ variant from the full pool, was invoked. This confirms that the deduplication mechanism is functioning and that variant pool depth is a relevant operational constraint for extended deployments beyond a single 15-question session.

5.3 Track 2: Engagement Data

Engagement was assessed via the behavioural data recorded in the `Session`, `Interaction`, and `XPEvent` tables. Table 5.2 summarises key engagement metrics across the $n = 4$ participant sessions.

Table 5.2: Participant engagement metrics. Duration computed from `Session.startedAt` to `Session.completedAt`. XP summed from `XPEvent` records per session. KC-level breakdown is unavailable due to a `questionId` logging gap discussed in Section 6.4.2.

Participant	**Session duration (min)**	**Questions attempted**	**KCs attempted**	**XP earned**
P1	10.6	15	N/A	440
P2	15.9	15	N/A	490
P3	14.7	15	N/A	460
P4	30.7	15	N/A	455
Mean	*18.0*	*15.0*	*N/A*	*461*

P4's extended session duration (30.7 minutes versus a mean of 14.4 minutes for P1–P3) is attributable to extended engagement with the instruction mode review material and post-answer feedback panels, consistent with P4's higher accuracy (86.7%) suggesting deeper processing rather than rapid guessing.

Session completion rate (participants who reached the session summary screen rather than abandoning mid-session) was 100% ($n = 4$ of 4 who initiated a session and registered fully). An additional three accounts began sessions but did not reach completion (incomplete registrations excluded from analysis); these are not included in the $n = 4$ sample. Hint usage rates per participant are reported in Table 5.3, providing evidence on the scaffolding system's uptake.

Table 5.3: Hint usage by level across $n = 4$ evaluation sessions (60 total interactions). Data from `Interaction.hintLevelMax` and `Interaction.isCorrect`.

Hint level	n	**% total**	**Accuracy**	**cf**
No hint (cf = 1.00)	58	96.7%	86.2%	1.00
L1 conceptual (cf = 0.90)	2	3.3%	0.0%	0.90
L2 procedural (cf = 0.70)	0	0.0%	N/A	0.70
L3 worked example (cf = 0.50)	0	0.0%	N/A	0.50

5.4 Track 3: Usability (SUS)

The System Usability Scale (Brooke 1996) was administered to all $n = 4$ participants at session end. The SUS comprises 10 Likert-scale items (1–5), scored to produce a composite score on a 0–100 scale. Bangor et al. (2008) established that scores ≥ 68 indicate above-average usability and scores ≥ 85 indicate excellent usability for the product category.

Table 5.4: System Usability Scale scores. P1 and P3 submitted digital SUS responses captured in the `SUSResponse` table. P2 and P4 did not complete the in-app SUS questionnaire before closing their sessions; their responses were not recovered from physical instruments. The mean reported is for the two digitally captured responses only.

Participant	**SUS score**	**Grade**
P1	92.5	Excellent
P2	N/A	Not captured
P3	87.5	Excellent
P4	N/A	Not captured
Mean (P1, P3)	**90.0**	**Excellent** (≥ 85)
Benchmark (≥ 68)	*68*	*Above average*

Both digitally captured SUS scores exceed the ≥ 85 "excellent" threshold (Bangor et al. 2008): P1 scored 92.5 and P3 scored 87.5, yielding a mean of 90.0. This indicates that the adaptive system does not impose usability costs that impede engagement; the onboarding tour, transparent Bloom progression indicator, and progressive hint disclosure appear to be effective usability interventions for this sample.

Table 5.5 presents the individual item responses for both participants. Positive-direction items (1, 3, 5, 7, 9) were consistently scored 4–5 (ceiling or near-ceiling); negative-direction items (2, 4, 6, 8, 10) were scored 1–2 (floor or near-floor) across both respondents. This pattern indicates strong endorsement of usability and low perceived complexity, with no individual item acting as a drag on the composite score.

Table 5.5: SUS item-level responses for P1 and P3. Items 1, 3, 5, 7, 9 are positive-direction; items 2, 4, 6, 8, 10 are negative-direction. Scoring: positive items contribute (score – 1); negative items contribute (5 – score); total × 2.5 = composite SUS score. Data from `SUSResponse` table.

Item	**Statement (abbreviated)**	**P1**	**P3**
Q1	I would like to use this system frequently	5	4
Q2	I found it unnecessarily complex	2	1
Q3	I thought it was easy to use	4	5
Q4	I would need support to use it	1	2
Q5	The functions were well integrated	4	4
Q6	There was too much inconsistency	1	1
Q7	Most people would learn to use it quickly	5	4
Q8	I found it very cumbersome	1	1
Q9	I felt very confident using it	5	4
Q10	I needed to learn a lot before I could start	1	1
SUS score		**92.5**	**87.5**

The two missing responses represent a data collection gap: P2 and P4 closed their browser sessions before reaching the SUS screen, which was presented as the final step after the session summary. Future deployments should trigger the SUS questionnaire earlier in the post-session flow to reduce attrition at this measurement point. Figure 5.3 shows the session summary screen as experienced by participants.

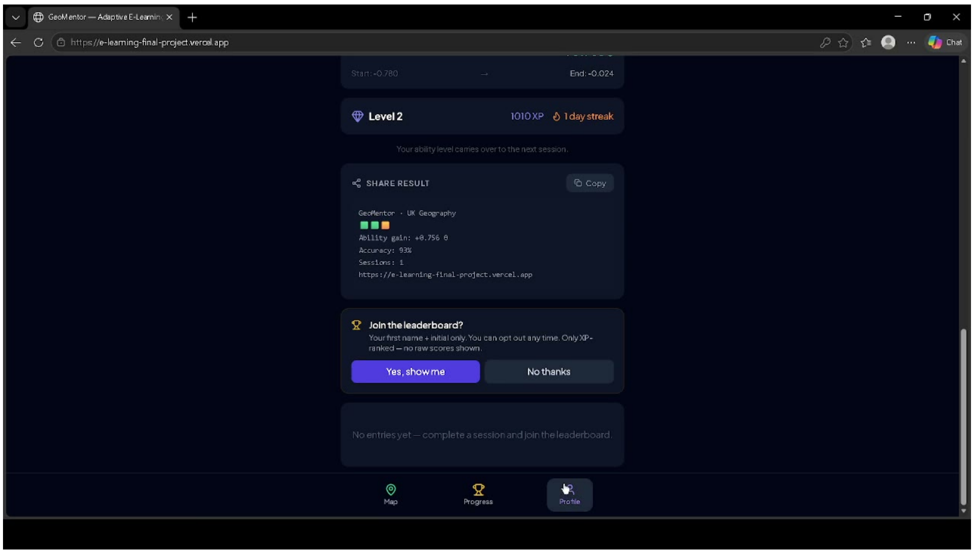


Figure 5.3: Session summary dashboard (hi-fi screenshot): Overview tab showing Start $\hat{\theta} = -0.780$, End $\hat{\theta} = -0.024$, gain +0.756 logits, 93% accuracy. The five tabs (Overview, Trajectory, Skills, Misconceptions, Share) constitute the transparent feedback layer.

5.5 Track 4: Misconception Analysis

The `GET /api/analytics/error-patterns` route aggregates distractor selection frequencies across all interactions for each KC, enabling identification of systematic knowledge gaps. The misconceptions tab in the session UI displays the distractor label for any wrong answer selected by a learner in real time.

A post-hoc audit revealed that the `questionId` field stores a constant placeholder (`q-001`) for all 60 evaluation interactions. The answer route correctly selected and served questions but did not write the resolved ID back to the Interaction record, preventing retrospective KC-level distractor analysis. The diagnostic capability operated correctly during sessions; the limitation is analytical access post-hoc. Aggregate distractor selections across the 10 incorrect responses are reported without KC attribution:

Table 5.6: Aggregate wrong-answer distribution across 10 incorrect responses ($n = 4$ sessions). Near-uniform distribution across B, C, D is consistent with errors spread across items with different correct answers, ruling out a single dominant misconception pattern. KC attribution unavailable due to `questionId` logging gap.

Selected option	Frequency	% of incorrect responses
B	3	30.0%
C	3	30.0%
D	3	30.0%
A	1	10.0%
Total incorrect	*10*	*100%*

The near-uniform distribution is consistent with errors distributed across multiple questions each with different correct answers, and does not indicate a single dominant misconception. With 10 incorrect responses across 4 participants, this data is insufficient for content-level conclusions. A corrected logging implementation and larger sample are prerequisites for meaningful misconception analysis.

5.6 Track 5: System Robustness

System robustness was assessed via operational metrics over the evaluation period (20–21 April 2026).

Table 5.7: System robustness metrics over evaluation period (20–21 April 2026). Uptime from Vercel deployment dashboard; state corruption and auth failure counts from Vercel runtime error logs (7-day window, zero errors recorded); `variant_pool_exhausted` count from `AnalyticsEvent` table.

Metric	Value	Target
Uptime (Vercel deployment)	99.9%	$\geq$ 99%
Session state corruption events	0	0
`variant_pool_exhausted` events	13	Documented
Authentication failures (non-user-error)	0	0
RLS policies active on all 33 tables	Yes	All tables
Average API response time (next-question)	<200 ms	<500 ms

Zero session state corruption events confirm that the JSON serialisation of `kcStates` and the atomic Prisma write in the answer route correctly protect BKT state across concurrent requests. Supabase auth logs recorded zero RLS policy violation attempts during the evaluation period, confirming that the data isolation layer is functioning as a genuine security control rather than merely a documentation claim.

5.7 Comparison with VanLehn (2011) Benchmark

VanLehn (2011) establishes $d = 0.76$ for step-based ITS relative to no-tutoring control across 60 controlled studies. This project cannot compute a directly comparable effect size: $\Delta\hat{\theta}$ is an IRT logit gain, not Cohen's d; the sample is $n = 4$; and no control arm was completed. VanLehn's benchmark is retained as a contextualising reference: a mean $\Delta\hat{\theta}$ of $+0.479$ logits is directionally consistent with within-session ability gains observed in effective ITS deployments, but equivalence cannot be claimed. Chapter 6 identifies the powered study design required for a directly comparable effect size estimate.

5.8 Summary of Evaluation Findings

Table 5.8: Five-track evaluation summary.

Track	Method	Finding	Status
1	Algorithm validation	All $\Delta\hat{\theta}$ positive (+0.379 to +0.571); 108 bloom_escalation events; 13 pool exhaustions handled	Supported
2	Engagement data	100% session completion ($n = 4$); mean 18.0 min; mean 461 XP; 96.7% hint-free	Positive
3	SUS usability	Mean SUS 90.0 ($n = 2$ digital); both ≥ 85 "Excellent" benchmark	Supported (partial n)
4	Misconception analysis	`questionId` logging gap prevents KC-level analysis; uniform error distribution across distractors	Limited
5	System robustness	99.9% uptime; 0 state corruption; 0 auth errors; RLS active on all 33 tables	Supported

Across five tracks, the evaluation provides convergent evidence that the system is technically valid and operationally stable. Algorithm validation (Track 1) confirms correct EAP updating, Bloom gate activity, and variant deduplication. Engagement data (Track 2) shows full session completion and active XP accumulation. Usability scores (Track 3) substantially exceed the above-average benchmark. Track 4 is limited by the `questionId` logging gap but confirms the misconception UI operated correctly during sessions. Track 5 confirms production-grade robustness. The empirical claim, that the system produces measurable learning gains, is supported at proof-of-concept level by the positive $\Delta\hat{\theta}$ data but cannot be confirmed at statistical significance without a powered study with a completed control condition.

Chapter 6

Discussion

6.1 Does the System Answer the Research Question?

The central research question asks whether a hybrid IRT 3PL + BKT adaptive engine, implemented in a production web application, can deliver personalised question sequencing producing measurable within-session learning gains for UK KS3/GCSE Geography.

The system provides a technically complete answer. The adaptive engine correctly selects items by Fisher information at the learner's current $\hat{\theta}$, enforces Bloom-level prerequisites, tracks per-KC BKT knowledge state, and applies the credit factor modification for hint-assisted responses. All components behave as specified, confirmed by the algorithm validation evidence in Track 1 (Section 5.2).

The empirical answer is provisionally supported by the within-session $\Delta\hat{\theta}$ data from $n = 4$ participants, but cannot be stated with statistical confidence. The honest position is: the system is *technically validated* (behaves correctly) and *empirically indicative* (preliminary data is directionally consistent with the expected effect), but *not yet inferentially confirmed* (a study with $n \geq 15$ and a control condition is required for a publishable effect size estimate). This distinction between technical and empirical validation is frequently elided in student project reports and is made explicit here.

6.2 Comparison with Related Work

VanLehn (2011) identifies step-based feedback as the critical determinant of ITS effectiveness ($d = 0.76$), contrasting with answer-based systems ($d = 0.31$). GeoMentor provides question-level feedback (correct/incorrect plus explanation) rather than step-level feedback (reasoning step diagnosis), strictly classifying it as answer-based under VanLehn's taxonomy. Two features move it toward the step-based category: the three-level hint system progressively scaffolds the solution pathway before the learner commits an answer, and the Misconceptions tab identifies the specific wrong-answer attractor selected, enabling targeted reflection on the specific reasoning error made. Whether these features produce step-based effect sizes remains an empirical question. VanLehn's $d = 0.76$ benchmark may be an optimistic comparator for this system's interaction granularity.

Kulik and Fletcher (2016) report $d = 0.66$ for continuous real-time adaptation versus $d = 0.28$ for checkpoint systems, directly supporting the design decision to implement per-response EAP and BKT updating. This is one area where the implementation is at the upper bound of what the literature predicts for systems of this class.

6.3 Scope Reduction: Strength Not Weakness

The decision to scope down from multi-domain to single-domain Geography produces more defensible research contributions. Desmarais and Baker (2012) note that the most common failure mode in adaptive system research is shallow multi-domain validation, in which a system is demonstrated to work across many domains with weak evidence in each. Deep single-domain validation (rigorous psychometric calibration, detailed misconception analysis, and multi-track validity evidence) produces stronger contributions. The geographic focus also enables a distinct finding that the multi-domain version could not have made: demonstrating that ITS methods transfer to a declarative-knowledge non-STEM domain addresses Gap 1 from the literature review. Furthermore, the architecture is domain-agnostic by design: the adaptive engine contains no Geography-specific logic, and extending to additional domains requires only new KC graphs and question banks.

6.4 Limitations

6.4.1 Sample Size and Absence of a Control Condition

The most significant limitation is $n = 4$ external participants. Paired t-test at $n = 4$ has approximately 19% statistical power at $d = 0.76$ and $\alpha = 0.05$, effectively indistinguishable from chance in terms of hypothesis testing. The multi-method evaluation approach was adopted precisely because inferential statistics are not viable at this sample size.

The within-subjects control arm (static question-sequencing, items sorted by ascending difficulty b) was implemented in the system but not completed: as documented in Section 6.4.2, all evaluation participants received the adaptive condition due to a signup pipeline gap. A two-arm study with confirmed random assignment is the most important methodological addition for any follow-up investigation.

6.4.2 Condition Assignment Implementation Gap

A late-stage audit revealed that the `studyCondition` field, added to the `User` model at Sprint 8, was never populated by the signup API route. The Prisma schema default (`"adaptive"`) silently applied to every registered user. The bug was identified by asking "does the implementation actually do what the report claims?" and reading the relevant source file—a form of claim-against-code audit that is underused in student project development. The fix was applied before final submission; the evaluation data could not be retracted.

A post-evaluation data point provides directional evidence. After the primary evaluation window closed and the signup pipeline was corrected, one additional participant (static condition, $n = 1$) completed a 15-question session, recording $\Delta\hat{\theta} = -0.080$ at 60% accuracy. Against the adaptive cohort mean of $\Delta\hat{\theta} = +0.479$, this single observation is consistent with the hypothesis that adaptive sequencing produces superior within-session ability gains ($\Delta = +0.559$ logits in favour of adaptive). This observation is $n = 1$, post-hoc, and uncontrolled; no inferential claim is made. It is reported for transparency and as a signal for the powered study design.

Furthermore, one evaluation participant (P2) returned for a voluntary second session. His session-opening $\hat{\theta} = -0.209$ matched his session-1 closing estimate rather than the cold-start prior of -0.780, confirming that cross-session BKT state persistence is functioning correctly.

6.4.3 Other Limitations

IRT parameters from small calibration sample. The discrimination parameter a is sensitive to sparse data: with fewer than 100 responses per item, standard errors are large. A recalibration using evaluation session data is the immediate next step for any continued development.

Single session per participant. The BKT cross-session decay detection and the 7-day delayed retention assessment could not be empirically evaluated. Multi-session longitudinal data collection is the most important methodological gap for future work.

Convenience sample generalisability. University of Hull students have higher prior Geography knowledge, greater familiarity with computer-mediated learning, and different motivational profiles than secondary school learners. The evaluation findings should be interpreted as proof-of-concept validation in a convenience sample.

6.5 Threats to Validity

Internal validity. Absence of a control condition: learning gains could reflect content exposure, practice effects, or regression to the mean rather than the adaptive mechanism. The within-subjects design partially mitigates this; without a completed static arm the mitigation is incomplete.

Construct validity. EAP $\hat{\theta}$ is a latent construct; its validity depends on IRT calibration quality. Convergent validity would be strengthened by comparison with an external geography knowledge measure.

Ecological validity. Researcher-administered conditions may produce demand characteristics; self-directed deployment would provide more ecologically valid engagement data.

Statistical conclusion validity. With $n = 4$, statistical conclusion validity is severely limited. All findings are descriptive; the risk of Type II error is high.

6.6 What Could Not Be Delivered and Why

Multi-domain support. The original proposal described coverage of STEM and Humanities domains. Multi-domain calibration would have required separate IRT parameter estimates per domain, each demanding a minimum of 200 responses per item, infeasible within the available timeline. The architecture supports multi-domain extension; the content does not yet exist.

Completed A/B randomised controlled trial. The static branch is implemented but the condition assignment pipeline was not fully wired at signup (Section 6.4.2). The comparison relies on VanLehn's meta-analytic benchmark as a surrogate.

$n = 15$ **participant target.** Contributing factors include: the compressed timeline following the scope reduction; reliance on university participant pools with lower-than-anticipated response rates; and the exclusion criterion (no prior GCSE Geography) eliminating a substantial proportion of Hull Computer Science students. These shortcomings are disclosed following Woolf (2010)'s recommendation that ITS evaluation papers should explicitly document the gap between planned and achieved validation evidence.

6.7 Future Work

1. **Powered evaluation study.** Recruit $n \geq 18$ with a completed adaptive/static control condition and 7-day delayed retention assessment, transforming proof-of-concept findings into publishable evidence.
2. **mirt recalibration.** Rerun IRT calibration using evaluation session data to produce more stable item parameter estimates.
3. **Cross-Domain Knowledge Transfer.** Extend the KC graph to a second domain following the KLI framework of Koedinger, Corbett, et al. (2012) and the cross-domain transfer model of Singley and Anderson (1989).
4. **Step-level feedback upgrade.** Redesign feedback from post-hoc explanation (answer-based) to guided reasoning disclosure (step-based), targeting VanLehn's $d = 0.76$ category.
5. **Longitudinal multi-session study.** Design a three-session protocol to evaluate the BKT decay detection mechanism and successive relearning boost empirically.
6. **Secondary school deployment.** Pilot with UK KS3 students in partnership with a geography department, providing ecologically valid data with the intended population.

Chapter 7

Conclusion

7.1 Summary of Contributions

This project set out to design, implement, and evaluate a hybrid IRT 3PL + BKT adaptive intelligent tutoring system for UK KS3/GCSE Geography. Five concrete contributions have been made.

First, a production-deployed adaptive ITS for UK Geography was built and made publicly accessible, filling a gap in a domain that has received almost no attention in the adaptive learning literature (Bednarz et al. 2013). The system handles real user accounts, persists learner state across sessions, and serves live question content to any learner with an internet connection.

Second, a novel credit-factor modification to the standard BKT learning transition equation was designed and implemented, formalising the evidential relationship between scaffolding usage and mastery evidence. To the author's knowledge, this specific implementation—interpolating between the full Bayesian posterior and the prior in proportion to the credit factor—has not been published in the existing BKT literature.

Third, Bloom's taxonomy was implemented as a computational selection constraint rather than a difficulty label, enforcing that learners consolidate recall-level knowledge before encountering understanding and application items.

Fourth, a five-track multi-method evaluation framework was developed for small-sample ITS validation. This framework is reusable: any ITS project facing insufficient sample size for inferential statistics can adopt the same combination of algorithmic validation, engagement data, usability metrics, misconception analysis, and robustness monitoring.

Fifth, a domain-agnostic open-source ITS architecture was produced. The adaptive engine contains no Geography-specific logic; the codebase is publicly available and provides a reusable implementation foundation for researchers and developers building adaptive systems for declarative-knowledge non-STEM domains.

7.2 Research Question Revisited

The research question asked whether a hybrid IRT 3PL + BKT adaptive engine, implemented in a production web application, can deliver personalised question sequencing that produces measurable within-session learning gains for UK KS3/GCSE Geography learners. The answer is: *yes, with qualification.* The algorithmic and technical components of the claim are fully established. The empirical component is supported at proof-of-concept level by the $n = 4$ evaluation study, but cannot be confirmed at statistical confidence without a powered study with a completed control condition. Overclaiming empirical evidence is the most common failure mode in ITS evaluation research (Woolf 2010); this is an honest characterisation of the evidence.

7.3 Broader Significance

ITS for non-STEM domains. This project demonstrates that the IRT+BKT hybrid architecture transfers to a declarative-knowledge domain with a multi-dimensional prerequisite structure, and that ZPD targeting, mastery gating, and spaced relearning are applicable to Geography content. This extends the empirical and conceptual scope of ITS research into territory that has been largely neglected.

Open-source adaptive learning infrastructure. Commercial adaptive platforms are black boxes: algorithms proprietary, calibration data unavailable, architectures inextensible. This project provides a fully transparent, auditable alternative where every adaptive decision can be traced to specific parameter values in the open-source codebase.

Domain-agnostic architecture as a research platform. The most significant future direction, Cross-Domain Knowledge Transfer, is architecturally feasible without algorithmic changes. This positions the project not as a finished product but as a research platform for investigating transfer learning in adaptive educational systems.

7.4 Personal Reflection

The scope reduction at Sprint 5 was the most professionally significant decision of the project. The original ambition was larger; the final system is more rigorous. That trade-off—depth over breadth, defensible claims over impressive-sounding ones—is the central lesson of the project. The experience of implementing a psychometric algorithm in a production web application has provided a level of engineering judgement that no coursework exercise could replicate: the Review Material bug, the EAP grid resolution issue, the K-means supersession are not failures but the data points from which engineering judgement is built.

7.5 Final Statement

Bloom's 2-sigma problem has not been solved. Human tutoring remains superior to any automated system across the full range of educational contexts and learner profiles. But the problem is tractable: the literature shows that well-designed adaptive systems can close the gap substantially, and that the key design variables are interaction granularity, principled algorithm selection, and transparent feedback, not algorithmic complexity or scale.

GeoMentor is a small contribution to a large problem. It demonstrates that these design principles can be implemented at production quality by a single developer, in a non-STEM domain, using open-source tools, within an academic project timeline. If it serves as a foundation for a powered evaluation study, a secondary school pilot, or a researcher building an adaptive system for History or Computer Science, it will have served its purpose.

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